



Out-of-class review and learner metacognition: Reciprocal effects in a blended learning course

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Abstract

Metacognition plays a crucial role in language learning, enabling learners to reflect on and adjust their strategies for more effective learning, particularly when reviewing for quizzes and exams. Technology can be a powerful aid in this context. This study examines the extent to which review-focused cognitive strategies, delivered using an online blended learning tool, support the metacognition of university-level English learners in Japan. Thirty learners in total participated in the study. They were enrolled in the teacher-researcher's thirty-week, first-year, compulsory English reading and writing course. The instruments employed were two learner reports, an end-of-course questionnaire, learner records from the tool's database, and the teacher-researcher's research journal. Findings indicated that: (1) the tool supported learners' metacognition by raising their awareness of gaps in their English-related linguistic knowledge, aiding their strategic planning for upcoming quizzes and exams, and providing opportunities for learners to explore new cognitive strategies; (2) many learners used their metacognitive skills to maximize their effort and select and use cognitive strategies with intentionality; and (3) the overall assignment had a positive impact on the test-anxiety of many learners. These findings confirm the potential of blended learning tools for fostering out-of-class review behaviors and learners' metacognitive development. Pedagogical implications are discussed and suggestions for future research are provided.

Keywords: learner metacognition; self-regulation; review; independent learning; blended learning, generative AI

Introduction

Helping language learners develop metacognition is essential, as it plays a central role in fostering self-regulation skills (Efklides, 2009; Oxford, 1990; Wenden, 1998; Xu, 2023; Zhang & Zhang, 2019). Despite its importance, metacognitive development is often overlooked in language classrooms, where instructional time is primarily devoted to linguistic content (Haukås, 2012, 2016; Wilson & Bai, 2010). Moreover, while some learners may expect their teachers to provide guidance about study strategies (Luk, 2013), others may have limited interest in out-of-class study, such as for reviewing (Lin, 2009). Given these challenges, there is a growing need to explore how technology can facilitate both structured review by learners while simultaneously fostering their metacognitive development.

Blended learning, which combines in-class instruction with technology-mediated independent study, has been widely adopted in second language education due to its benefits for L2 learning (Adair-Hauck et al., 2013; Young, 2008). It is through such a format that teachers could provide learners with the kind of structured review and focus on metacognitive development that is needed. Further, recent research related to the integration of artificial intelligence (AI) (AITwijri & Alghizzi, 2024; Liang et al., 2021; Woo & Choi, 2021; Yang & Kyun, 2022) indicates that the technology could be drawn on to facilitate this. However, while AI has been shown to support language learning, the extent to which AI-driven review-focused activities could contribute to learners' metacognitive development remains an open question.

To address this gap, I drew on an existing online blended learning tool for English-language teaching and learning and added functionality that would allow me to provide structured review activities. These would require learners to use review-focused cognitive strategies (see Oxford, 1990). I then designed a study that would encourage learners in one of my courses to complete the activities out of class to prepare for quizzes and exams in the course. For the purposes of this study, I have termed these independent review activities (henceforth, IRAs). They are termed “independent” since learners do not complete them in class and because they independently choose how many and which of the activities to complete. One included IRA is doing simple review, where learners see feedback that the teacher has entered into the tool's database during class time and click on an item to get an explanation about it from generative AI. Another is learners having a text-based conversation with AI about a topic from their course. Non-AI-based activities are also included—for example, learners re-reading textbook pages or contributing voice-recorded responses to topic-based questions for their classmates to listen to and potentially respond to.

In the context of the study, beyond helping learners review linguistic content, it was expected that the activities would help learners achieve growth in their review-focused metacognition, and based on the current design of the IRAs, specifically their metacognitive knowledge. This was because the learners would be encouraged to complete multiple IRAs before quizzes and exams in the course, and through so doing, potentially further build their awareness of gaps in their linguistic knowledge, the kinds of activities they could complete to review effectively, and what activity types they thought worked best for them as individuals. Any growth achieved would be considered technology-mediated metacognitive growth. This means that learners' interaction with the tool would serve as the primary driver of their metacognitive development, rather than direct metacognitive training provided by the teacher, for example.

By integrating IRAs—some of which incorporate AI—into a blended learning course, this study examines how structured review activities contribute to learners' metacognitive development. While extensive research has examined technology-mediated metacognition, most of it has focused on metacognitive strategies (see Monika & Devi, V, 2022 for a systematic review).

Relevant literature (e.g., Huang et al., 2023) indicates that more research is needed that focuses on technology-mediated metacognitive knowledge (however, see Supriyono et al., 2020) and efforts to develop it (however, see e.g., Bangkom & Sukavatee, 2021; Lee, 2015). This study addresses that gap. In so doing, the study contributes to the growing body of research on technology-mediated metacognition and its role in second language learning.

The Literature Review

Metacognition

Learner self-regulation has long been recognized as essential for the development of language-learning skills, and metacognition is essential during the process of self-regulation (Efklides, 2009; Oxford, 1990; Wenden, 1998; Zhang & Zhang, 2019). Self-regulation refers to “self-generated thoughts, feelings, and behaviors that are oriented to attaining goals” (Zimmerman, 2002, p. 65). Metacognition involves awareness of, monitoring, and regulating one’s cognitive processes (Flavell, 1979; Zimmerman, 2002). Metacognition consists of three components: metacognitive knowledge, metacognitive experiences, and metacognitive strategies (Flavell, 1979). These components will now be discussed separately.

Metacognitive Knowledge. This is defined as knowledge about one’s learning (Wenden, 1987). It is a constellation of beliefs and knowledge (Zhang & Zhang, 2019) that learners acquire “formally or informally, deliberately or incidentally” (Wenden, 1998, p. 516). This knowledge forms the foundation for self-regulated learning and influences how learners approach tasks and strategies. It can sometimes be inaccurate (Efklides, 2009; Zhang & Zhang, 2019), leading to ineffective learning approaches. For example, a learner who believes they learn best by memorizing vocabulary lists may neglect more effective strategies like using words in context, which inevitably impacts their progress.

There are three types of metacognitive knowledge: person, task, and strategy. Person knowledge refers to learners’ awareness of factors that influence learning, including their own. It can relate to their own abilities and their beliefs about their abilities. Task knowledge refers to what learners know about the purpose of a task, the nature of the task, and task demands. Strategy knowledge refers to “general knowledge about what strategies are, why they are useful, and specific knowledge about when and how to use them” (Wenden, 1998, p. 519). For example, a learner preparing for a listening test might recognize that they struggle with fast speech (person knowledge), understand that the task requires identifying key details (task knowledge), and therefore decide to practice with slowed-down recordings before gradually increasing speed (strategy knowledge).

Metacognitive Experiences. Flavell (1979) defines these as “any conscious cognitive or affective experience that accompany and pertain to any intellectual enterprise” (p. 906). They involve “awareness, unexpected awareness, thoughts, intuitions, perceptions, feelings, and self-judgements of oneself as a cognisor during problem solving and task completion” (Tarricone, 2011, p. 130). These experiences lead to new ones and prompt revision of learners’ metacognitive knowledge. For example, a learner who suddenly realizes mid-sentence that they are misusing verb tenses may commit this to memory and become more attentive to the grammar point during future speaking activities.

Metacognitive Strategies. According to Wenden (1998), these are “general skills through which learners manage, direct, regulate, [and] guide their learning, i.e., planning, monitoring, and evaluation” (p. 519). The deployment of these can be referred to as self-regulation (Wenden, 1998). Learners must apply the strategies consciously and purposively (Efklides, 2009). They help the learner monitor their cognitive progress, and they can activate both cognitive and

metacognitive strategies (Flavell, 1979). For instance, a learner studying for an exam may plan their study sessions (planning), check their understanding by summarizing key points (monitoring), and reflect on which techniques were most effective (evaluation).

Metacognition in ELT

In the context of ELT, research has shown that learner metacognition facilitates L2 learning (Teng, 2019; Teng & Zhang, 2016; Vandergrift et al., 2006). Furthermore, while learners have been shown to be generally aware of strategies, such knowledge (or, in other words, awareness) alone does not always translate into effective strategy use (Zhang & Goh, 2006). That said, the more metacognitive awareness learners have, the more they use strategies (Nosratinia et al., 2014). Given this, it has been recommended that learners receive strategy training (Wenden, 2002). Anderson (2002) proposed a metacognitive strategy model consisting of five steps: (1) preparing and planning for learning; (2) selecting and using learning strategies; (3) monitoring strategy use; (4) orchestrating various strategies; and (5) evaluating strategy use and learning. Anderson (2002) notes that these steps are not strictly linear, as multiple processes often occur simultaneously. Importantly, targeted efforts to further train learners regarding strategies have been shown to yield positive results (Peñuela, 2018; Plonsky, 2011).

Metacognition, motivation, and self-efficacy are closely interconnected factors that reinforce one another in complex ways. Studies have shown the positive link between metacognition, motivation, and self-efficacy (Nosratinia et al., 2014; Vandergrift, 2005). Bandura (1997) defines self-efficacy as “beliefs in one’s capabilities to organize and execute the courses of action required to produce given attainments” (p. 3). Nosratinia et al. (2014) demonstrated that learners’ self-efficacy positively predicts both their metacognitive awareness and strategy use. Regarding motivation, Vandergrift (2005) found that the more self-determined a learner’s motivation was, the more they engaged in metacognition. Alternatively, Dörnyei and Skehan (2003) argue that strategy use may sustain learners’ motivation. This indicates that strategy use and motivation may have a reciprocal relationship.

The Cognitive Basis of Review in SLA

Both theoretical perspectives and empirical research in the field of Second Language Acquisition (SLA) underscore the importance of review for language development. First, the Noticing Hypothesis (Schmidt, 2001; Schmidt & Frota, 1986) asserts that conscious attention to linguistic features is essential, supported by longitudinal evidence from Schmidt and Frota (1986). Second, Associative Learning Theory (Ellis & Wulff, 2014) highlights how repeated exposure strengthens memory, with studies showing that repeated vocabulary encounters lead to greater lexical gains (Elgort, 2011; Webb, 2007). Third, Skill Acquisition Theory (DeKeyser, 2015) and the Declarative/Procedural Model (Ullman, 2015) emphasize that practice transitions explicit knowledge into automatic use. Suzuki and Sunada (2018) found that repeated, meaningful language use aids automatization. Fourth, the Output Hypothesis (Swain, 1985; Swain & Lapkin, 1995) argues that producing language refines linguistic knowledge, with Leeson (2008) demonstrating that pushed output enhances accuracy. Fifth, the Interaction Hypothesis (Long, 1983, 1996) posits that negotiation of meaning aids acquisition, a claim supported by Mackey (1999), who found that learners engaging in interactional modifications showed significant linguistic gains. Sixth, depth of processing (Craik & Lockhart, 1972; see also Izumi, 2002) and integrative processing (Graf & Schacter, 1989) have been linked to better retention. Finally, Sociocultural Theory (Vygotsky, 1978) emphasizes the role of social interaction, with Alzubi, Nazim, and Ahamad (2024) showing that collaborative learning improves EFL proficiency. Collectively, the above indicates that effective review should promote noticing, repeated

exposure, meaningful practice, production, interaction, and deep processing, all of which contribute to long-term retention and language development.

Blended Learning

Blended learning combines synchronous interactive study (typically face-to-face) with asynchronous (individual) study (usually online) (A. King, 2016). The asynchronous component often includes e-learning, where learners engage with software or online tools to support their learning objectives (Sasaki & Takeuchi, 2024; Yamamoto, 2021).

The online blended learning tool used in this study integrates AI, a growing focus in language learning research (AITwijri & Alghizzi, 2024; Liang et al., 2021; Woo & Choi, 2021; Yang & Kyun, 2022). Studies have demonstrated the benefits of both blended learning (Adair-Hauck et al., 2013; Young, 2008) and AI (Teng & Yang, 2022; Woo & Choi, 2021) for L2 acquisition.

Blended learning environments require learners to take responsibility for their learning (Adigun et al., 2024). General-education research shows that learners with stronger metacognitive skills apply more effective strategies in online settings (Bernacki, Aguilar, & Byrnes, 2011) and that self-efficacy shapes metacognitive regulation (ElSayad, 2024; Kassab et al., 2015; Stephen & Rockinson-Szapkiw, 2021).

In second language learning, metacognitive strategies enhance learner success in blended courses (Monika & Devi. V, 2022). However, learners benefit from support, as perceived teacher guidance predicts both self-efficacy and self-regulation (Rui & Liu, 2023). Metacognition has also been linked to greater engagement in blended learning (Saraff et al., 2020; Sasaki & Takeuchi, 2023, 2024).

Blended Learning, Review by Learners, and Metacognition in ELT

Research in ELT shows that language learners possess pre-existing metacognitive knowledge (Goh, 1997) and that targeted training can help refine it (Hauck, 2005). Learners also demonstrate technology-related metacognitive awareness, such as recognizing the value of specific online tools (Naseri & Motallebzadeh, 2016; Supriyono et al., 2020) or the value of specific strategies, like reading strategies (Sholeh et al., 2022). The online blended learning tool used in this study aims to support learners in developing review-focused metacognitive knowledge through the independent review activities (IRAs). When engaging in these activities, learners draw on their existing knowledge and strategies (Bandura, 1997; Zimmerman, 2000) and may also refine them through metacognitive experiences (Flavell, 1979). That is, they may achieve technology-mediated metacognitive growth.

The Operationalization of Learner Metacognition in this Study

In this study, metacognition is operationalized as any psychological or behavioral indicators of learners' awareness, monitoring, and regulation of cognitive processes. Metacognitive knowledge is operationalized as learners' knowledge and/or beliefs about the English language, gaps in their English linguistic knowledge, effective review approaches, and strategies which work best for them. Metacognitive strategies are operationalized as actions taken to manage the review process. Technology-mediated metacognitive growth is operationalized as increases in metacognitive knowledge or strategy use.

Methodology

This study employed a mixed-methods approach, integrating both quantitative analysis of learner questionnaire data and data generated through learner use of the online blended learning tool, and qualitative insights drawn from learners' reports and the teacher-researcher's journal.

Research Questions

The current study examines learners' completion of the independent review activities (IRAs) provided by the tool. Such activities included re-reading textbook pages, making voice recordings to share with classmates, engaging in conversations (with peers, family, or AI), and using online resources to explore class topics more deeply.

The study sought to answer the following research questions:

1. What role did learner metacognition play in their completion of the independent review activities?
2. How did the independent review activities support the learners' metacognition?

The Context and Participants

The study was conducted at a private university in Shizuoka, Japan. The participants ($n = 30$) were first-year learners in a compulsory reading and writing course of mine in the International Relations faculty. The course ran for fifteen weeks, and we met twice a week on Tuesday and Friday mornings. The learners, aged 18 to 19, were International Relations majors. Twenty learners were male and ten were female. The learners were divided into two classes: the 1st-period class (A2 level) and the 2nd-period class (A2+ level) based on the Common European Framework of Reference for Languages (CEFR) (Council of Europe, 2021). Both classes were categorized as "Basic User", with the 2nd-period class having slightly higher proficiency. Variation in proficiency level among learners in the same class was expected and observed, however. We used a print textbook that focused on four skills (primarily reading and writing) related to different countries and cultures. Learners completed homework activities at home, and in class, I reviewed their work, addressed difficulties, and engaged them in communication activities related to the homework topics.

Ethical Considerations

Ethical approval for this study was obtained from the university prior to data collection. Participation was voluntary, and learners provided informed consent after being briefed on the study's purpose, data collection methods, and confidentiality measures. To protect participant anonymity, unique codes were assigned, and all identifying records were securely stored and later destroyed in accordance with ethical guidelines.

The Online Blended Learning Tool

The online blended learning tool was *Global Communication Stars* (Stewart, 2025) (see Appendix B for representative images). The design of the tool was informed by literature in SLA and teacher development (e.g., Blin, 2016; Borg, 2015; Brown & Lee, 2015; Chapelle, 1998; Dörnyei, 2001; Garrett, 2009; Nation & Macalister, 2010). I self-developed the tool so that language teachers would have an online system that was able to provide a range of useful functionality for blended English-language learning courses and for supporting their ongoing professional development.

The learners' side of the tool includes features for blended learning, with generative AI integrated to provide automated feedback and learner-centered review activities. The teacher's side supports real-time instruction with functions like feedback entry and display, AI-powered vocabulary translations and linguistic examples, tallying of points during language-learning games, presentation of strategies, and analytics on learner performance. Further details of the tool's functionality that is relevant to this study are outlined below.

The IRA types completed by learners during the current study are described in Table 1¹. To design the activities, I drew on a number of theories (see the far-right column in the table and the key in the footer).

Table 1 *The IRA types*

Type	Details	Relevant Theories
Do simple review	The learner could see the feedback that I provided in class and could click on an item to get an explanation from generative AI in both English and their native language.	1
Re-read textbook pages	The learner would choose one or more textbook pages to re-read.	2, 3
Make a voice recording	The learner would make a voice recording to check their pronunciation of words that they had studied in class. No feedback was provided, so the learner would self-assess based on practice that we had done in class.	1, 3, 4
Contribute to the “About Us” section (See Figures B6 and B7 in Appendix B)	The learner would voice record a response to one or more of the questions listed, which all related to topics covered in the class. They could both record their own answers and listen to their classmates’ and record responses to them.	1, 3, 4, 5, 7, 8
Have a conversation with someone out of class	The learner would have a conversation in English about a course topic outside of class with their friend or a family member.	1, 3, 4, 5, 6, 7, 8
Use the Internet to study more deeply about class topic	The learner would choose an aspect of a previous class that they would like to learn more about and access the Internet to read a text or watch a video about it, ideally in English. They chose what to read or watch themselves.	3, 4, 7
Have a conversation with generative AI (See Figures B4 and B5 in Appendix B)	The learner would choose a topic and have an interaction with generative AI. The topics were from their textbook. The prompts had been written by me in advance and were stored in the system database.	1, 3, 4, 5, 6, 7

A valid concern is whether the IRAs, as designed, can effectively create the conditions required by the underlying theories to facilitate linguistic growth and if learners will complete the IRAs as intended to fully benefit from them. This concern is acknowledged, but as the current study

¹ Key: 1 - The Noticing Hypothesis, 2 - Associative Learning Theory, 3 - Skill Acquisition Theory, 4 - The Declarative/Procedural Model, 5 - The Output Hypothesis, 6 - The Interaction Hypothesis, 7 - Depth of processing and integrative processing, 8 - Sociocultural Theory (Vygotsky, 1978).

focuses on the initial implementation of the IRAs, refinement of the IRAs will be addressed in future research.

Figure 1 presents the IRA choice section. Learners chose which IRAs to complete from this section. Only those IRAs appearing in the table above were included in the study (the others were added later as a result of the study). Figure 2 presents what the learners saw after they clicked the button for “Do simple review”.

Instruments and Data Collection

This study employed both quantitative and qualitative data collection methods to examine learners’ engagement with independent review activities (IRAs) and their metacognition. Quantitative data was gathered through Likert-scale survey responses and database records, while qualitative data was obtained from open-ended survey responses, learner reflections, and my research journal. Five instruments were used for the study.

Learner Report 1. This was completed by the learners in class three. It listed eight questions related to learner motivation and review habits (see Appendix A). The instrument was constructed by the teacher-researcher. Data from this instrument was used to answer RQ 1.

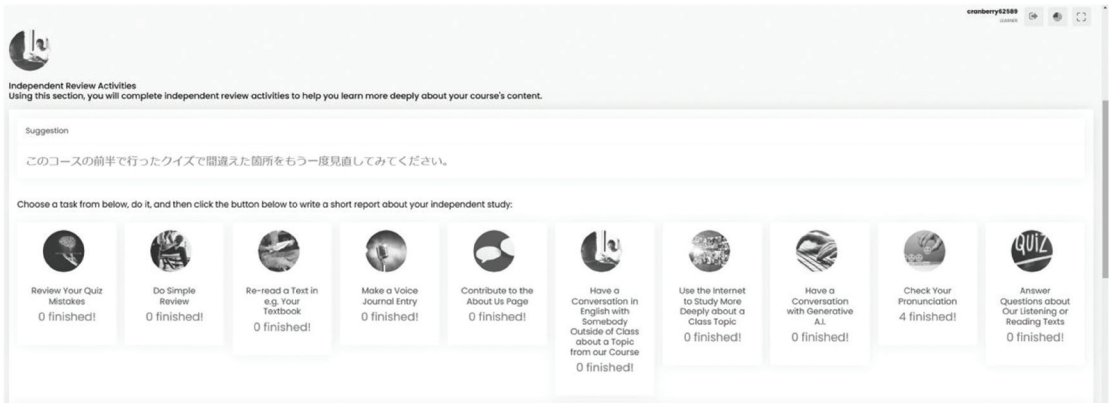


Figure 1 The learners’ side of the tool (the “Independent Review Activities” choice section)



Figure 2 The learners’ side of the tool (the “Do Simple Review” section)

For questions 1–4, the data type was ordinal. For question 3, a six-point Likert scale was used since I wanted a definitive indication of their perception of their own competence related to review. Questions 5–8 required open-ended responses. The learners were given a Japanese-language version of the instrument. They were asked to answer in either English or Japanese. By allowing learners to respond in Japanese, it was hoped that their answers would be more detailed given that they would not be constrained by their English proficiency. When analyzing the learners' responses, I found that this was the case. Overall, few learners chose to write in English. Some wrote in both Japanese and English in places.

Learner Report 2. This was completed by the learners in the class immediately after the mid-term exam to capture their reflections while their experiences with the IRAs were still recent. The instrument was constructed by the teacher-researcher. The instrument listed four questions related to their completion of the IRAs (see Appendix A). Data from this instrument was used to answer RQs 1 and 2. For all questions, open-ended responses were required. The learners were given a Japanese-language version of this instrument. They were asked to answer in either English or Japanese.

The End-Of-Course Questionnaire. The questionnaire used in the current study was completed by the learners in class twenty-nine immediately before the final exam. The instrument was constructed by the teacher-researcher based on the adapted version of the Motivated Strategies for Learning Questionnaire (MSLQ; Pintrich & De Groot, 1990) created by Sasaki and Takeuchi (2024). It contained seventeen items about test anxiety, motivation/engagement, and metacognition (see Appendix A). For all items, the data type was ordinal. A seven-point Likert scale was used this time to provide learners with the option to indicate uncertainty, ensuring they were not forced into agreement or disagreement. The learners were given a Japanese-language version of this instrument. Data from this instrument was used to answer RQs 1 and 2.

Learner Records in the Database. The learner records in the online blended learning tool database were used to determine which IRAs the learners had completed before taking the mid-term exam. There were a total of 441 records. The database row for each IRA done contained data in five fields: *l_id* (*learner_id*), *class_code*, *activity_type* (this ranged from 0 to 6; for example, the number for activity type “Do simple review” was 2), *evidence*, and *status* (i.e., as set by the teacher: “accepted”, “not accepted”, “add more details”, or “correct the mistakes”). Data from this instrument was used to answer RQs 1 and 2.

The Teacher-Researcher's Research Journal. This journal (see Dörnyei, 2007) was used to log any important events related to conducting the study and record any observations and reflections that related to the learners' use of the tool and/or their metacognition. Data from this instrument was used to answer RQs 1 and 2.

Data Analysis

Learner Reports 1 and 2. Immediately after collecting the data, I compiled it into a spreadsheet and cleaned it to ensure accuracy. During the term, I engaged in an iterative process of reading and reflecting on the data at multiple points, but particularly at the beginning. This process aimed to identify patterns in learner responses that could shed light on their potential and ongoing engagement with the independent review activities (IRAs) and indicate any further support that the learners might need. At the end of term, for the quantitative data (items 1–4), I calculated descriptive statistics (mean, mode, and standard deviation). I also generated frequency distributions. This analysis was conducted using JASP (The JASP Team, 2018). For items 5–8 (open-ended questions), I conducted thematic analysis (Saldaña, 2009). I applied structural coding to determine the frequency for each common learner response.

A similar qualitative approach was taken for Learner Report 2 during the term. At the end of term, for items 2 and 4, I applied both descriptive and structural coding to derive themes and to determine the frequency for each common learner response. I did not formally analyze data for items 1 and 3.

The End-Of-Course Questionnaire. A quantitative approach was used for this instrument. The data was first compiled and cleaned, and descriptive statistics, including mean, mode, and standard deviation, were calculated. Additionally, correlational analysis was conducted to explore the relationship between two specific variables to answer the second research question. Full details of this analysis are provided in the Findings section.

Learner Records in the Database. A quantitative analysis was conducted to calculate the completion frequency of each IRA type. This data was used to rank learners’ preferences for different IRAs and to examine patterns in the combinations of IRA types completed in one sitting (e.g., types 1, 2, and 7 as a set). These analyses offered insights into learners’ study habits and preferences, contributing to a better understanding of their engagement with the IRAs and their metacognition.

The Teacher-Researcher’s Research Journal. For this dataset, I applied descriptive coding to identify themes related to the two research questions.

Triangulation of Data. The data from each research instrument was analyzed separately for each research question. However, to ensure triangulation, all data was then compared to identify overarching patterns.

The Procedure

Before the commencement of the course, I reviewed relevant literature and reflected on learners’ metacognition, considering their context, self-efficacy, and motivation to assess their readiness for the independent review activities (IRAs). Based on this, I decided to emphasize the purpose of the IRAs, provide basic training, and ensure learners reflected on their completion of them at the end of the term.

In the first class, learners were informed about the study, signed consent forms, and created anonymous accounts for the tool. The tool’s interface was available in both English and Japanese. From class two onwards, I used the teacher’s side of the tool to enter and display feedback on common mistakes (see Figure 3), which learners would be able to access during completion of one specific IRA from class three.

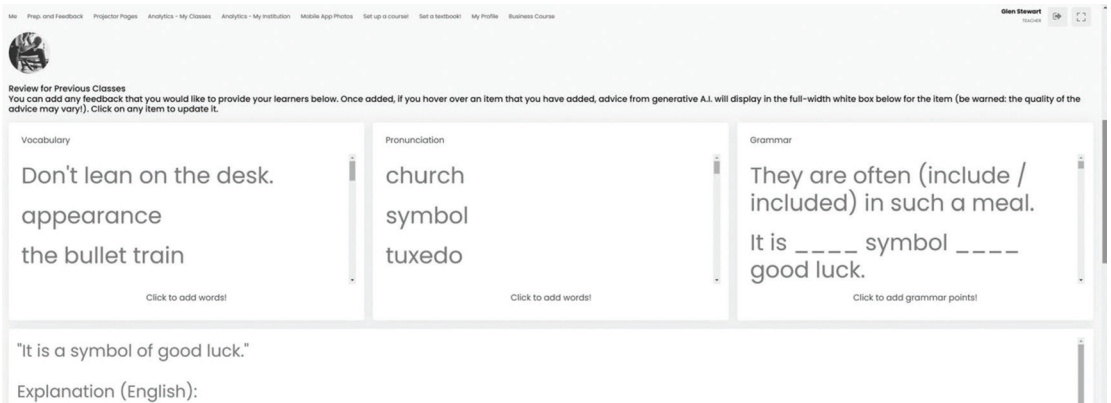


Figure 3 The teacher’s side of the tool (the “Review for Previous Classes” section)

In class three, learners completed Learner Report 1 and received training, including an explanation of each IRA type. To complete an IRA, they were asked to select one, complete it, provide “evidence” of completion (e.g., why they chose it, what English they used, what English they learned, etc.), and submit their response to the system. I encouraged learners to complete some IRAs requiring deeper-level processing (i.e., using specific linguistic elements to have a conversation with generative AI). After this class, the learners completed at least three IRAs at home to prepare for the first quiz.

In class five, I reviewed the number of IRAs each learner had completed and assessed the “evidence” they provided (see Figure 4). Entries were marked as “Accepted” or categorized as “Not accepted” (if repeated, unclear, or too brief), “Write more details” (if insufficient), or “Correct the mistakes” (if errors were present). I verified the evidence as needed and provided feedback before the learners took the quiz, which assessed vocabulary and grammar from prior classes. This cycle—quiz announcement, IRA completion, feedback, and quiz taking—was repeated in classes eight/nine.

Just before class twelve, I encouraged the learners to complete more IRAs to prepare for the mid-term exam. In class fifteen, they took that exam. In class sixteen, after giving feedback about the exam, I asked the learners to complete Learner Report 2.

The cycle for classes three/five and eight/nine was repeated in classes twenty/twenty-one. In class twenty-six, the learners were again encouraged to complete as many IRAs as possible to prepare for the final exam. As part of this, I suggested reflecting on which IRAs had been most helpful to them to date and completing more of those. In class twenty-nine, before the final exam, the learners completed the End-of-Course Questionnaire, scheduled beforehand for practical reasons.

In class thirty, I clarified unclear responses from three learners and updated the data. I then asked all learners to answer the following question in a short written reflection: “By completing the IRAs, have you learned anything new about quiz and exam revision?” Learners then volunteered to share their reflections with the class (and I asked other specific learners to read theirs) and I recorded representative ones in my research journal. Next, I shared a summary of the findings of the study and my interpretations, conducted member checking to confirm the accuracy of my interpretations overall, and provided a bullet-point list of advice related to learner metacognition and review for quizzes and exams.

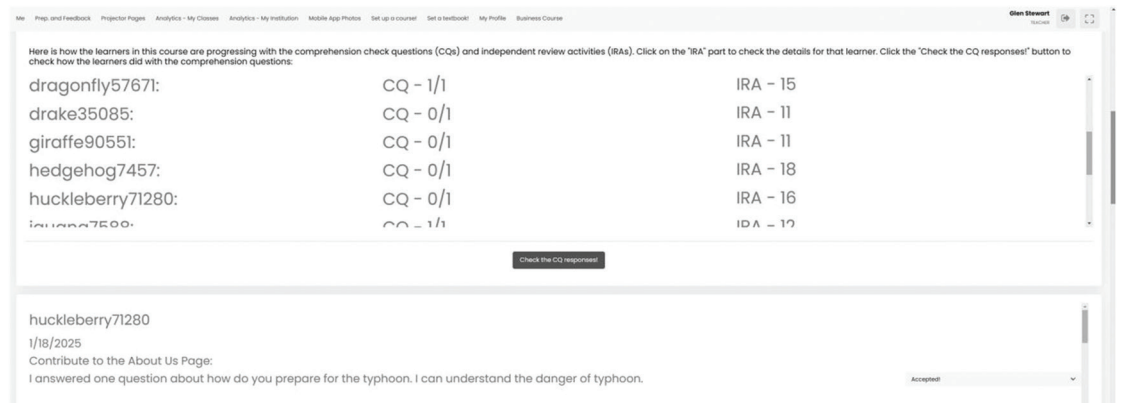


Figure 4 The teacher’s side of the tool (the “Learner Work” section)

Positionality Statement

A. King (2024) states that “the goal of positionality statements is to understand not so much the researcher’s developmental processes, but the research development processes” (p. 5). Since the researcher is inherently part of the data collection process (Bourke, 2014), it is important to acknowledge my role as the developer of the online blended learning tool. While I believed in its usefulness, I encouraged honest learner feedback about it and modified it, as needed. To maintain objectivity, I also self-monitored throughout the research process, ensuring that findings accurately reflected learner data.

Findings

The overall goal of having the learners complete the independent review activities (IRAs) was to provide an opportunity for them to use cognitive strategies within the context of the course to review for course-related quizzes and exams. The findings focus on how learners’ metacognition impacted their use of the cognitive strategies and how their metacognition was impacted as a result.

1. What role did learner metacognition play in their completion of the independent review activities?

To investigate this, I examined the number and types of IRAs learners chose, the combinations of IRAs they completed, and their self-reported reflections. The following tables summarize key findings from database records, learner reports, and their questionnaire responses.

Table 2 lists all independent review activities (IRAs) completed by the learners during the term and the frequency for each.

Table 2 The IRA types chosen by the learners

Type	Frequency
Do simple review	184
Re-read textbook pages	127
Have a conversation with generative AI	52
Use the Internet to study more deeply about a class topic	33
Have a conversation with someone out of class	24
Contribute to the “About Us” section	19
Make a voice recording	15

This indicates that the most popular activity types were doing simple review and re-reading textbook pages. Table 3 presents the frequency of combinations of activity types that the learners completed before the first quiz (i.e., before class five).²

We can see from Table 3 that the activity “Re-read textbook pages” is commonly paired with the activity “Do simple review”. These two appear to be a core combination for many learners

² Key: (1) Do simple review; (2) Re-read textbook pages; (3) Make a voice recording; (4) Contribute to the “About Us” section; (5) Have a conversation with someone out of class; (6) Use the Internet to study more deeply about class topic; and (7) Have a conversation with generative AI.

Table 3 *The combinations of IRAs chosen by learners to prepare for Quiz 1*

IRA Combination	Frequency
1, 2, 7	4
1, 2, 4	3
1, 2	2
1, 2, 6	2
1, 2, 3	2
1, 5, 7	2
1, 4, 7	1
1, 5, 6	1
1, 2, 3, 6	1
1, 7	1

Table 4 *Excerpts from the results of learner report 2 – How they used the IRAs to review*

Learner	Comment
TF1_5	I used them to review vocabulary and grammar that would be included in the mid-term exam.
TF2_2	I used them to review grammar that we had covered in class.
TF2_4	I looked at the explanations of vocabulary and grammar and confirmed what I had not memorized yet.
TF2_16	I used them to review vocabulary, grammar, and pronunciation that I learned in class. This is because the IRAs make it easy to understand what I need to review. I was able to review right away after class, so I was able to understand the class well.

when reviewing for the first quiz. The activity “Have a conversation with generative AI” often appears with those two activities. This may mean that it is a fun third activity to complete along with more traditional review activities. Analysis of the activities completed after the second quiz (which was done in class nine) and before the mid-term exam (which was done in class fifteen) indicates that the individual activity frequencies and the frequencies of activity combinations were very similar to those in Tables 2 and 3.

Table 4 presents a selection of learners’ responses to questions in Learner Report 2. These responses indicate how learners used the IRAs to review for the quizzes and the mid-term exam.

Learner TF2_16 appears to be relatively high in both metacognitive knowledge and skills as well as self-efficacy. This was evidenced by her behavior in class, her performance on several course assessment items, and the fact that she completed as many IRAs as any other learner.

On Learner Report 1, a majority of learners indicated knowing “some” (67%) or “not much” (21%) about reviewing. Further, when asked if they felt good at reviewing, there was variability in the data, with 33% choosing “2” or “3” (/6) and 45% choosing “4”. Overall, this indicates that

learners lacked confidence in their ability to review. Thematic analysis for item 7 on the report indicated that “re-read the textbook”, “review vocabulary”, and “look up things” (e.g., words) were the most common review strategies for learners.

Comments by several learners on Learner Report 2 indicated that they felt regret about not completing more IRAs prior to taking the exam (see Table 5).

Table 5 *Excerpts from the results of learner report 2 – Regrets*

Learner	Comment
TF1_10	I did not use them a lot. I should use them a lot to improve my English.
TF2_9	I did not get a good score because I did not review enough.
TF2_13	I did not use it much. If I had, my mid-term exam score might have been higher.
TF1_1	I do not study enough. I will do the IRAs more.

Thematic analysis of item 6 on Learner Report 1 revealed that the most common reasons learners cited for not reviewing generally were “no time”, “sleepiness”, and “work”. These factors may help explain why the above learners were unable to complete more of the IRAs.

Tables 2, 3, and 4 indicate that many learners drew on their metacognition to select and use strategies that they perceived to be helpful for preparing for the quizzes and the exam. To do this, they would have needed to draw on their awareness of their strengths and weaknesses as it pertained to the course material. Additionally, the opportunity to reflect after the exam afforded the learners metacognitive experiences which indicated to them the positive or negative consequences of their efforts. Providing such an opportunity is therefore essential to the overall process (in this case, it was provided outside of their use of the tool using one of the research instruments). In the case of learner regrets, that they perceived themselves to not have reviewed enough through completing IRAs may indicate a lack of metacognitive self-regulation.

When learners completed the End-Of-Course Questionnaire in class twenty-nine, they responded to several items designed to assess what metacognitive strategies they used when completing the IRAs. These items were included to capture key aspects of metacognition: self-regulation; degree of intentionality and planning in their approach; and monitoring and evaluation of the utility of the IRAs and making adjustments accordingly. Table 6 below summarizes data regarding this.

Table 6 *Learner metacognition during completion of the IRAs*

Report Item	Mean	Mode	Standard Deviation
I did my best when doing the IRAs.	5.12	6.00	1.333
I did only the minimum amount of work necessary to complete each IRA.	4.80	4.00	1.41
I chose which IRAs to do randomly.	3.64	5.00	1.80
I had a specific reason for choosing each IRA that I did.	5.36	6.00	1.20
I stopped using an IRA if I did not think it was useful.	3.52	4.00	1.45

The learners generally felt that they put effort into their IRAs (80% chose a score of “5” or higher (/7)) rather than on doing just the minimum. More learners chose IRAs with intentionality (73% chose a score of “5” or higher) rather than randomly³. Selecting activities with intentionality and abandoning use of activities that are not useful are metacognitive strategies. The lower tendency to abandon activities indicates either a lack of metacognition or that the IRAs were, for the most part, useful.

Most learners completed between 9 and 18 IRAs in total, but distinct performance clusters emerged. High performers completed between 15 and 18 IRAs, moderate performers completed 11 to 14, and low performers completed 10 or fewer. One learner in the low-performing cluster, TF1_2, chose self-ratings which indicated high engagement with the course, low enjoyment of the overall IRA assignment (he chose a “3” (/7); 75% chose a score of “5” or higher), high test-anxiety, and low metacognition during completion of the IRAs. This learner completed a total of 9 IRAs. Approximately 92.86% of learners completed more IRAs than this learner. Contrast that learner with one in the high-performing cluster, learner TF1_5. She chose self-ratings which indicated medium-level engagement with the course, low enjoyment of the overall IRA assignment (she chose a “1”), high test-anxiety, and low metacognition during completion of the IRAs—except for the two items related to choice of IRA. This learner indicated not choosing randomly (choosing “1”) and instead with intentionality (choosing “7”). She completed a total of 18 IRAs, which is more than approximately 85.71% of learners in the class.

I sought to investigate the above using correlational analysis. To test for normality, I first conducted a Shapiro-Wilk test, which yielded a p-value of 0.095, suggesting that the data did not significantly deviate from a normal distribution. Since one of the variables was measured on an ordinal Likert scale (1–7), I conducted a Spearman’s rank correlation to examine the relationship between the two variables: the number of IRAs completed and the learners’ tendency to choose activities randomly. The analysis revealed a significant negative correlation ($\rho = -0.569$, $p = 0.002$). This indicates that learners who selected activities in a more strategic manner tended to complete a greater number of IRAs, whereas those who chose activities randomly completed fewer IRAs.

What the overall findings for this first research question indicate is that many learners selected the IRAs with intentionality and self-regulated when completing them. Those that did not tended to complete fewer IRAs overall. Further, it is unclear if learners evaluated strategies after using them.

2. How did the independent review activities support the learners’ metacognition?

To investigate this, I analyzed learners’ reflections on their use of IRAs, their perceptions of how these activities supported their review, and their reported experiences with test anxiety. The following tables summarize key findings from learner reports, questionnaire responses, and in-class reflections.

Immediately after the mid-term exam, the learners were asked to complete Learner Report 2. This instrument asked the learners to reflect about their completion of the IRAs to review for the exam. The results for this instrument indicate that one way in which the IRAs supported

³ However, low internal consistency was found between the items measuring random selection and intentional selection ($\alpha = 0.298$) and this suggests that learners may not have viewed these as strict opposites. Though the difference in the means is clear, the high modes for both items and high standard deviation for random selection could indicate that some learners believed they were making intentional choices overall while still engaging in random selection at times.

learners' metacognition was by raising their awareness of gaps in their English-related linguistic knowledge. This is evidenced by comments such as the following. Learner TF2_4 wrote:

I looked at the explanations of vocabulary and grammar and confirmed what I had not memorized yet.

Completing the "Do simple review" IRA helped learners to identify these gaps (a metacognitive experience). While the IRA itself involved a cognitive strategy (reviewing feedback and AI explanations), engaging with the feedback reflected self-monitoring of one's progress and potential metacognitive knowledge increase, particularly within the person category.

Another way in which the IRAs supported learners' metacognition was by aiding strategic planning. Strategic planning, a key component of metacognition, involves learners making intentional decisions about how to approach their review based on their goals and needs. Below is a learner response to the question of how they used the IRAs to review. This is from learner TF1_5:

I used them to review vocabulary and grammar that would be included in the mid-term exam.

This comment indicates that she used the IRAs with a clear purpose: targeting content that was directly relevant to the upcoming assessment item. This level of intentionality shows an ability to identify priorities and focus learner efforts on areas of importance.

A further way in which the IRAs supported the learners' metacognition was that they provided them with a chance to try out one or more new cognitive strategies using technology (e.g., having a text-based conversation with generative AI about a topic from the course). As can be seen in Table 2 in the previous section, there were 52 instances of learners having a conversation with generative AI.

Just before the final exam, the learners were asked to complete the End-Of-Course Questionnaire. The questionnaire included two items related to test anxiety: a general item - "I worry about tests." (mean = 5.60, mode = 7.00, standard deviation = 1.61) and one that referred specifically to the IRAs - "Doing the IRAs helped to reduce my worries about the quizzes and exams." (mean = 4.96, mode = 6.00, standard deviation = 1.31). Overall, 64% of learners indicated that the IRAs had helped them to reduce their test anxiety.

The results of the reflecting that learners did in class thirty about their completion of the overall IRA assignment reveals that several learners recognized the value of various IRAs: TF1_6 found frequent review beneficial; TF1_10 saw Internet-based review as effective; TF2_9 viewed AI conversations as useful; TF2_14 stated that she had come to realize that there were many ways to review, and that she had not been bored when reviewing during the course; and TF2_16 stated that she had learned the importance of doing simple review (e.g., of vocabulary, pronunciation, and grammar). This provides evidence that the overall assignment had a positive impact on the metacognitive knowledge of at least some of the learners.

It is important to also note the following result. Thematic analysis for item 2 on Learner Report 2 indicated confusion by two learners about the connection between the IRAs and the mid-term exam. This is from learner TF2_7:

I was able to check grammar in detail, but not spelling, so the result was not good.

This points to an assumption that there would be a strong connection between the IRAs and the mid-term exam.

What the overall findings for this second research question indicate is that the IRAs supported learners' metacognition by enhancing their awareness of their own knowledge gaps, aiding in their strategic planning, and encouraging exploration of new strategies involving technology. In addition, the IRAs appeared to help many learners reduce test anxiety. However, the findings also indicate confusion about the goal of the IRAs among at least two learners.

Discussion

The findings of this study indicate that using an online blended learning tool for English-language learners to complete independent review activities (IRAs) for review purposes supported learners' metacognition, while their pre-existing metacognition also played a largely facilitative role in their use of the tool. I will now discuss the findings.

Technology-Mediated Metacognitive Growth

According to Zhang and Zhang (2013), "learners' metacognition has to undergo continuous change and adaptation" (p. 114). This was an impetus for the development of the IRAs and conducting the current study. Early in the study, I had the expectation that learners' completion of the IRAs would result in an expansion of their metacognitive knowledge. Increases in both person knowledge and strategy knowledge appear to have been achieved. For example, as reported in the findings, there were 52 instances of learners having a conversation with generative AI. In one way, this relates to the person category of metacognitive knowledge. This category relates to learners' awareness of factors that influence learning, including their own. Whatever the reason for the popularity of this activity type (perhaps its novelty; Flavell, 1979), both during and after conversing with generative AI, the learners would have had metacognitive experiences. It would appear that, at least in some cases, those experiences further built person knowledge.

As Wenden (1998) points out, learners can acquire metacognitive knowledge "deliberately or incidentally" (p. 516). In the case of the overall IRA assignment, the goal was for that acquisition of knowledge to be somewhat deliberate. That is, I was hoping that the learners would come to understand the value of review and the value of the different IRA types for engaging in it. This also appears to have been achieved to some extent. This represents review-focused technology-mediated metacognitive growth. This finding is a positive given the need for teachers to provide more metacognitive training to learners (Haukås, 2012, 2016; Luk, 2013; Wilson & Bai, 2010). That the majority of the learners found the overall IRA assignment enjoyable is a positive given that some university-/technical-school-level learners may be reluctant to engage in out-of-class review without some kind of structured assignment (Lin, 2009).

Test Anxiety and Test Washback

In this study, learners completing the IRAs appeared to have a beneficial impact on their perceptions of their performance on the mid-term exam and their test-anxiety. That is a positive. However, the findings also indicate that some learners approached completion of the activities with incorrect assumptions. It was necessary for me to point out on several occasions that there would not be a completely direct connection between the IRAs and the quizzes and exams. As Efklides (2009) and Zhang and Zhang (2019) point out, learners' metacognitive knowledge can be inaccurate. In this case, learners' assumptions reflected a misalignment between their metacognitive knowledge (i.e., task knowledge) and the actual learning and teaching process

(i.e., that teachers try to avoid negative test washback; that is, a situation in which learners study only what is needed to do well on a test).

Metacognition and Engagement

In their study, Sasaki and Takeuchi (2024) found that learner engagement with e-learning out-of-class may have been mediated by metacognitive ability. They posed the question: “Which components of the students’ metacognition increased their engagement in e-learning, and how?” (pp. 10–11). In the case of the current study, the findings suggest a similar link between learner engagement and metacognition. A significant negative correlation was found between the number of IRAs completed and the random selection of activities, indicating that learners who engaged in more strategic activity selection tended to complete more IRAs. Further, learner TF1_5 reported only moderate engagement with the course and limited engagement with the overall IRA assignment but reported choosing the IRAs with intentionality. This may indicate more of a role for metacognition than motivation in this learner’s case. One answer to the above question may be, therefore, “learners choosing which units or activities to do with intentionality (i.e., choosing based on one’s own needs and goals)”. In an earlier study, Sasaki and Takeuchi (2023) found that strategy knowledge influenced engagement. The IRAs in the current study likely activated learners’ strategy knowledge, with the overarching strategy being “choosing review activities for quiz and exam preparation”. The dominance of traditional review strategies (e.g., re-reading textbook pages) may suggest that learners relied on familiar, established ones, indicating an impact of strategy knowledge on engagement. However, this also raises the question of whether they actively evaluated alternative activities beyond having conversations with AI or if their choices were primarily driven by perceived ease of completion.

Metacognition and Self-Efficacy

Previous research has shown that self-efficacy influences metacognitive regulation (ElSayad, 2024; Kassab et al., 2015; Nosratinia et al., 2014; Stephen & Rockinson-Szapkiw, 2021). Schunk (1989) explains that self-efficacy influences the choice and direction of behavior, assuming adequate skills, positive outcome expectations, and valued outcomes. I perceived learner TF2_16 to be high in self-efficacy. This was based on my perception of her behavior during class time. For example, she was always one of the first learners to raise her hand and offer an answer. More objective measures indicated that she was high in metacognition. This may have facilitated both her completion of the IRAs and her comparatively superior performance on most assessment items. This provides further evidence for the influence of self-efficacy on learner metacognition.

Conclusion

This study examined how 30 first-year English learners in Japan using an online blended learning tool for quiz and exam review impacted, and was impacted by, their metacognition. The findings suggest that the review activities enhanced metacognition by increasing awareness of knowledge gaps, aiding strategic planning, and offering opportunities to try new cognitive strategies. Most learners also reported reduced test anxiety. However, while many engaged purposefully, others participated less intentionally, suggesting that the tool’s effectiveness depended on learners’ metacognitive self-regulatory ability.

These findings have several implications for language teachers and those designing similar tools. First, incorporating structured review assignments within a blended learning environment using cognitive strategies can enhance learners’ metacognitive knowledge and strategy use while also reducing test anxiety. Second, teachers could consider having learners complete

such activities within the context of ongoing peer coaching to sustain engagement and deepen language processing through learner interaction. Third, blended learning tools should include metacognitive support, such as prompts for planning, monitoring, and evaluating activities (Bannert et al., 2009), particularly if such guidance is not provided elsewhere. Intelligent conversational agents could assist with this (Song & Kim, 2020). Given that teacher support influences self-efficacy and self-regulation (Rui & Liu, 2023), individualized guidance is crucial. Fourth, tailoring interventions to different learner profiles could provide necessary scaffolding. Finally, a greater variety of activities is needed, including those that promote accurate self-assessment (Dörnyei, 2001; Vandergrift & Tafaghodtari, 2010; Wang, 2015) and AI-driven quizzing, as repeated retrieval enhances retention more effectively than passive review (Karpicke & Roediger, 2008).

While this study offers valuable insights, several limitations should be considered. First, accurately verifying IRA completion was challenging since learners provided their own “evidence”, which could have been fabricated. To mitigate this, I closely monitored activity completion and followed up in class. Second, the End-Of-Course Questionnaire relied on learner self-reporting, which may be influenced by social desirability bias (Chamot, 2004). To counter this, learners were reminded that honesty was crucial and that responses would not affect their grades. Third, while the negative correlation between random selection and IRA completion is notable, correlation does not imply causation (Dörnyei, 2007). External factors could also influence both behaviors, such as learners with strong strategic planning skills completing fewer IRAs due to time constraints. Finally, the small, relatively homogeneous sample (e.g., mostly motivated learners) limits the generalizability of the findings.

Despite the limitations of the study, there are several promising avenues for future research stemming from this study. One area could focus on identifying the optimal mix of review activities to offer learners, ensuring that the activities align with different learner profiles and preferences. Another direction could involve examining the longer-term impact of completing review activities on learners’ linguistic performance in their courses. A future study might also delve into clarifying what makes learners choose traditional review strategies (e.g., re-reading textbook pages) over more interactive ones and how to encourage more exploration of the latter.

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Appendix A

The Learner Reports and the End-of-Course Questionnaire

Learner Report 1 Items

1. How motivated are you regarding English study?
 1. Very motivated
 2. Somewhat motivated
 3. Not very motivated
 4. Not motivated at all
2. How much do you know about reviewing for English quizzes and tests?
 1. A lot
 2. Some
 3. Not much
 4. Nothing
3. Are you good at reviewing for English quizzes and tests?
(Not at all) 1 .. 2 .. 3 .. 4 .. 5 .. 6 (Very much so)
4. How often do you review before English quizzes and tests?
 1. For all classes
 2. For this class and other classes
 3. For this class only
 4. For no classes
5. If you review, how long do you review for? (e.g. one hour)
6. What things stop you from reviewing? Provide at least 3 (e.g. a lack of time).
7. What are useful tactics/activities for reviewing for quizzes and tests? Provide at least 3 (e.g. re-reading a textbook page).
8. Imagine that you are going to re-read a textbook page in order to review for a quiz. What steps would you follow?

Learner Report 2 Items

1. How did you use the IRAs to review for the mid-term exam?
2. What impact did the IRAs have on your mid-term exam result?
3. Now that you have done the mid-term exam, how will you use the IRAs differently in the second half of the term?
4. What problems did you have? Do you need more training?

The End-Of-Course Questionnaire

1.	I worry about tests.	NT 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 T
2.	I feel an uneasy, upset feeling when I take a test.	NT 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 T
3.	I think that what we learned in this course was interesting.	NT 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 T
4.	What we learned was useful for me to know.	NT 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 T
5.	I think that I will be able to use what I learned in this course in other courses.	NT 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 T
6.	I had trouble avoiding distractions when doing the textbook homework.	NT 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 T
7.	When doing my textbook homework for this course, I just did the required sections and then closed the book.	NT 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 T
8.	When reading a text in the textbook, I stopped every so often to go over what I was reading.	NT 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 T
9.	I thought about what I learned after finishing my textbook homework.	NT 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 T
10.	I generally enjoyed doing the IRAs (e.g., “Do simple review”, “Have a conversation with AI”, etc.).	NT 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 T
11.	I found doing the IRAs to be boring or bothersome.	NT 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 T
12.	I did my best when doing the IRAs.	NT 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 T
13.	I did only the minimum amount of work necessary to complete each IRA.	NT 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 T
14.	I chose which IRAs to do randomly.	NT 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 T
15.	I had a specific reason for choosing each IRA that I did.	NT 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 T
16.	I stopped using an IRA if I did not think it was useful.	NT 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 T
17.	Doing the IRAs (e.g., “Do simple review”, “Have a conversation with AI”, etc.) helped to reduce my worries about the quizzes and exams.	NT 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 T

Note. NT denotes “not true for me” and T denotes “true for me”.

Appendix B

Representative images of the online blended learning tool

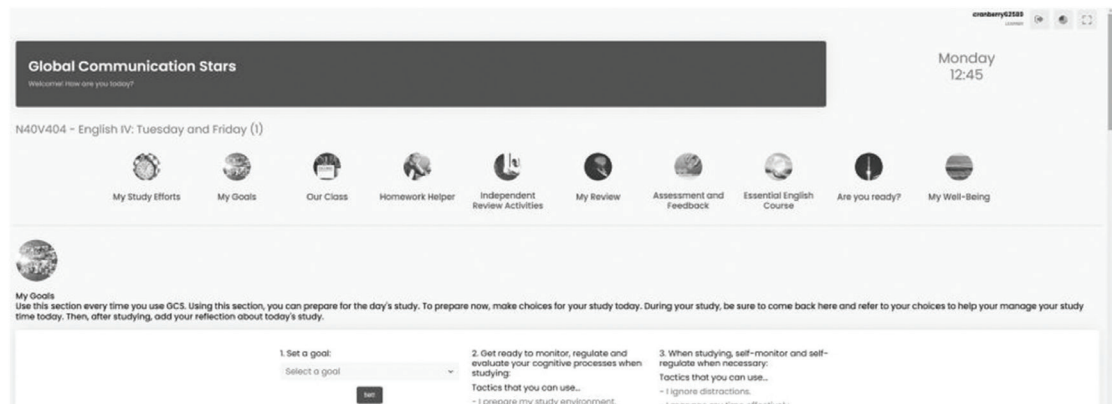


Figure B1 The learners’ side of the tool (the “My Goals” section)

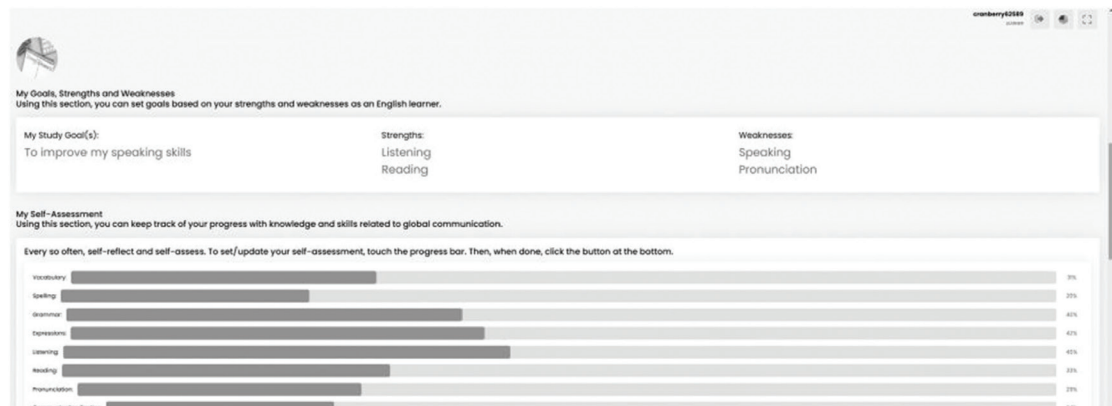


Figure B2 The learners’ side of the tool (the “My Self-Assessment” section)

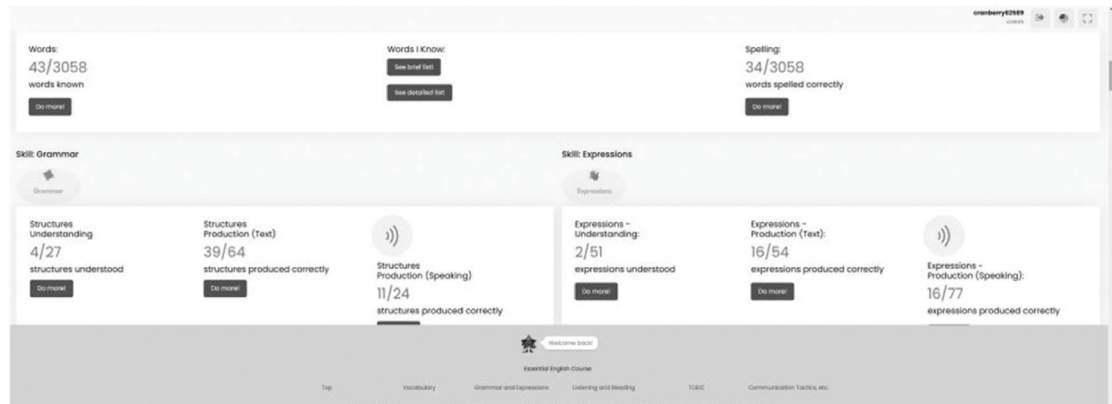


Figure B3 The learners’ side of the tool (the “Essential English Course” section)

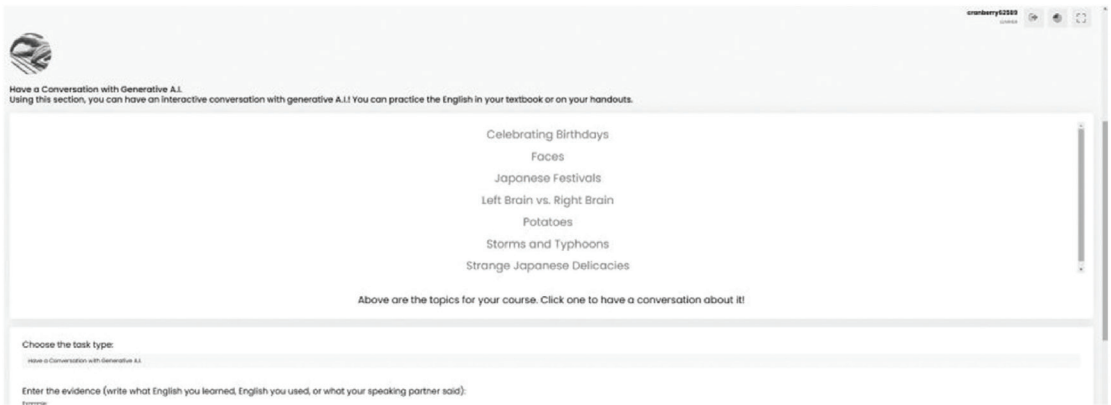


Figure B4 *The learners’ side of the tool (the “Have a Conversation with Generative AI” section)*

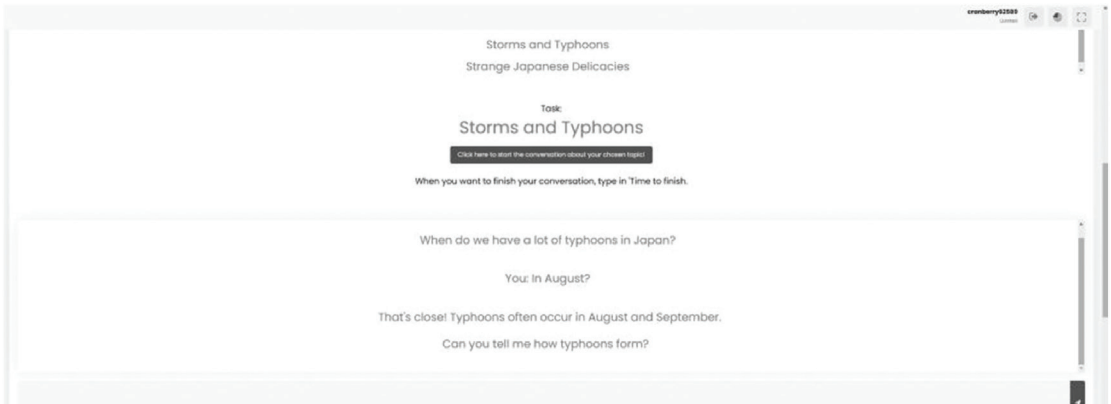


Figure B5 *The learners’ side of the tool (the “Have a Conversation with Generative AI” section)*

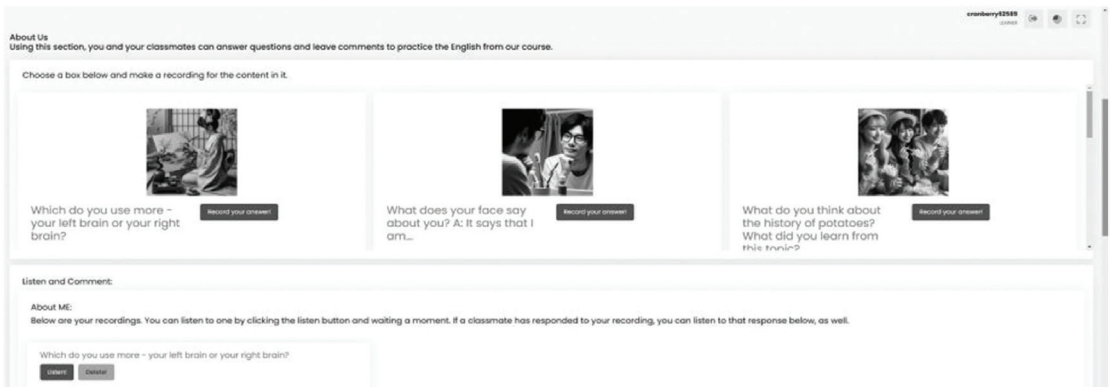


Figure B6 *The learners’ side of the tool (the “Contribute to the ‘About Us’” section)*

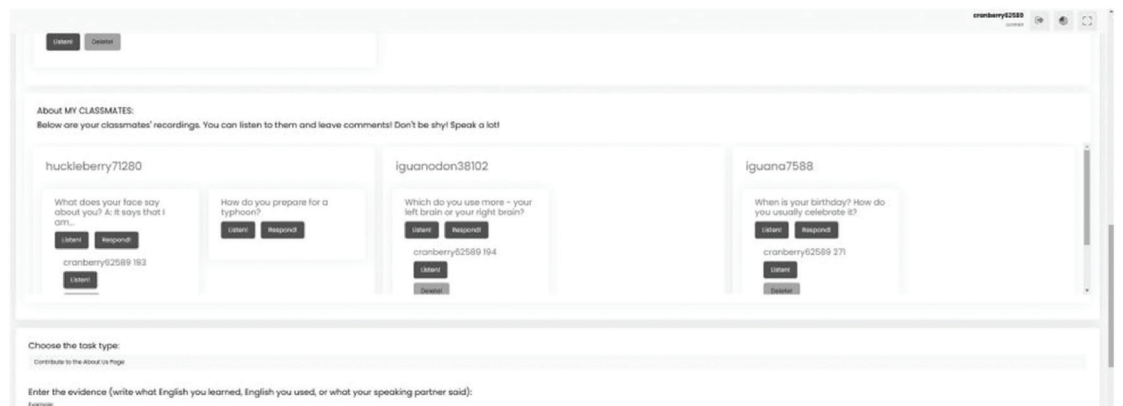


Figure B7 *The learners’ side of the tool (the “Contribute to the ‘About Us’” section)*

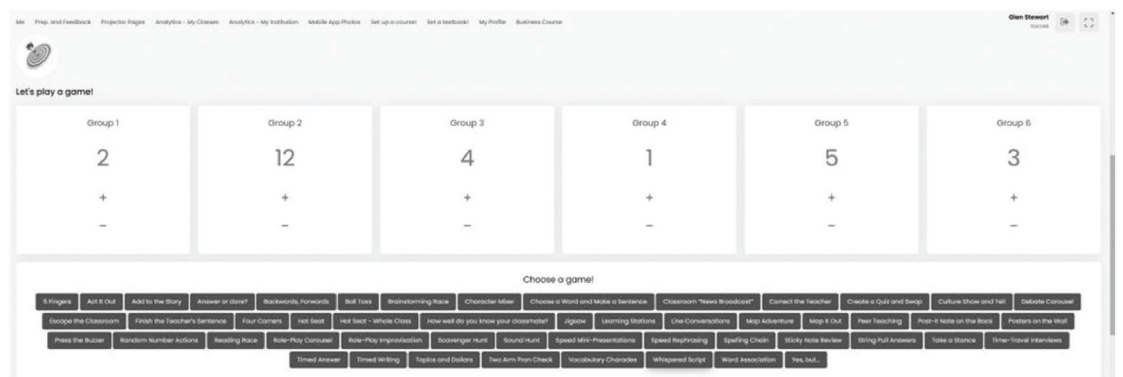


Figure B8 *The teacher’s side of the tool (the “Let’s play a game!” section)*

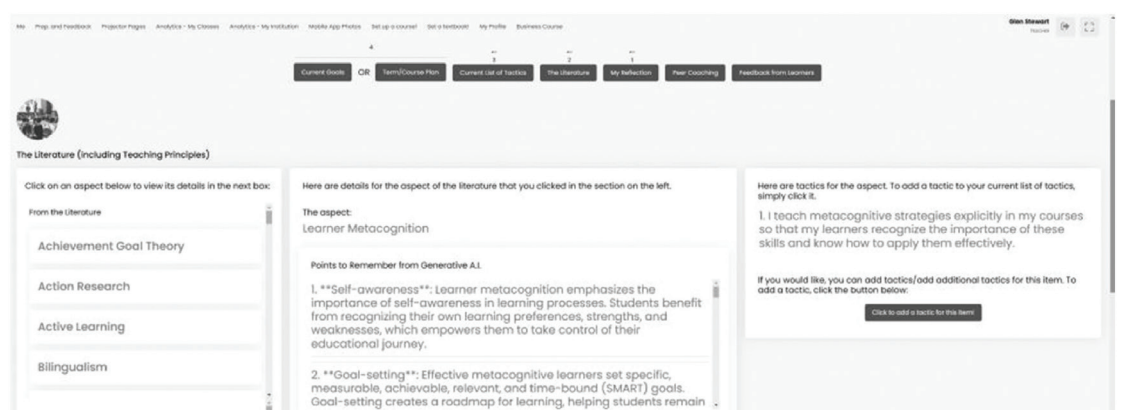


Figure B9 *The teacher’s side of the tool (the “The Literature” section for language teacher professional development)*

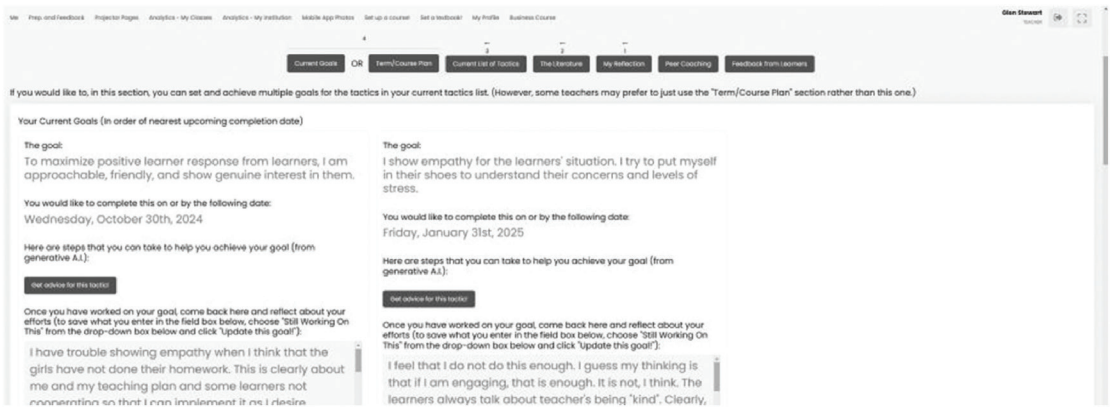


Figure B10 The teacher’s side of the tool (the “Current Goals” section for language teacher professional development)